Research Article

Multi-State Modeling After Bone Marrow Transplantation: A Continuous-Time Markov Chain Approach

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Abstract

This study implements a time-homogeneous Continuous-Time Markov Chain (CTMC) framework to model the longitudinal clinical pathways of patients following allogeneic bone marrow transplantation (BMT). Utilizing the `ebmt4` registry comprising 2,279 leukemia patients from the `mstate` package in `R`, we construct a 6-state multi-state topology encompassing four transient phases (Transplant, Recovery, Adverse Event, and Combined Rec+AE) and two competing absorbing states (Relapse and Death). Transition intensities for the infinitesimal generator matrix $\mathbf{Q}$ are estimated via parametric Maximum Likelihood Estimation (MLE), and macro-level cohort trajectories are resolved across short-term, one-year, and asymptotic landmarks using matrix exponential evaluations $\exp(\mathbf{Q}t)$. Empirical results indicate that the baseline leaving rate from the initial transplant state yields an expected sojourn time of 467.92 days, with a slightly higher initial intensity toward complications ($q_{13} = 0.000996$) over pure recovery ($q_{12} = 0.000862$). Asymptotically ($t \to \infty$), the system converges to an ultimate terminal absorption allocation of 44.14% for Relapse and 55.86% for Death. Compared to conventional univariate survival curves, this multi-state approach provides significantly higher prognostic differentiation by successfully capturing risk trajectories within intermediate transient states without informational loss due to non-informative censoring.

Keywords: *Continuous-Time Markov Chain, Multi-State Models, Infinitesimal Generator Matrix, Matrix Exponential, Bone Marrow Transplantation*

1. INTRODUCTION

Hematological malignancies, particularly leukemia, continue to pose a substantial global health and economic burden, demanding aggressive and sophisticated therapeutic interventions [15]. While the clinical maturation of allogeneic bone marrow transplantation (BMT) has fundamentally revolutionized leukemia treatment protocols [2, 4], the post-transplantation recovery trajectory is rarely linear and is heavily characterized by highly dynamic intermediate clinical phases. Patients do not simply transition from health to death; instead, they navigate a complex network of interleaved physiological events, such as graft-versus-host disease (GvHD), delayed platelet recovery, and the perpetual threat of disease recurrence based on clinical registry guidelines [19]. These clinical phases do not occur in isolation but function as a series of time-dependent competing risks, making the accurate prognostic tracking of post-BMT cohorts a critical necessity in modern biostatistics, mathematical statistics, and actuarial science [8, 13, 17, 18].

Conventional survival analysis techniques, such as the classical Kaplan-Meier product-limit estimator and the Cox proportional hazards model, possess fundamental structural limitations when handling these multidirectional clinical pathways [7]. Because traditional frameworks are inherently bivariate, they are strictly restricted to modeling a single, binary transition from an initial state to a single terminal endpoint, thereby rendering them inadequate for branching multi-state systems. To resolve this methodological constraint, multi-state modeling via a Continuous-Time Markov Chain (CTMC) framework is adopted [6, 12]. This stochastic framework explicitly maps the longitudinal trajectory of individuals through a finite discrete state space over continuous time [5]. By utilizing an infinitesimal generator matrix $\mathbf{Q}$ to capture localized transition intensities, the system leverages the Kolmogorov forward differential equations to resolve macro-level transient state occupancy probabilities over time [1, 9].

Although advanced multi-state techniques have become a standard benchmark internationally for evaluating clinical registries [11], their rigorous application within academic research in Indonesia remains markedly sparse, with the majority of domestic epidemiological literature still bound to conventional univariate models [16]. This study directly bridges this methodological gap by systematically executing a time-homogeneous CTMC analysis on the comprehensive `ebmt4` blood and marrow transplantation dataset, consisting of 2,279 leukemia patients [10]. By enforcing an explicit constant intensity constraint, this paper provides a robust first-order approximation of the system [3], unlocking precise, interpretable metrics including expected clinical sojourn times and long-term asymptotic absorption allocations that enhance the prognostic utility of standard medical data.

1.1. Problem Formulation

Based on the outlined background and structural constraints of the clinical registry, this research formulates the following problems:

1. How to construct a mathematical 6-state transition architecture that accurately enscapulates the dynamic, interleaved clinical trajectories of leukemia patients post-BMT?
2. How to estimate the constant transition intensities of the infinitesimal generator matrix $\mathbf{Q}$ using parametric Maximum Likelihood Estimation (MLE) under a time-homogeneous assumption?
3. To what extent does the time-homogeneous CTMC framework unlock advanced prognostic indicators—specifically explicit milestone probability matrices, expected sojourn times, and asymptotic absorption allocations—compared to conventional bivariate survival approaches?

1.2. Research Objectives

The primary objectives of this research are defined as follows:

1. To design a unidirectional 6-state transition graph and restructure the comprehensive longitudinal data into an exposure-at-risk long format compatible with multi-state processing.
2. To estimate the complete infinitesimal generator matrix $\mathbf{Q}$ by evaluating empirical transition events against cumulative person-days exposure via parametric maximum likelihood estimators.
3. To compute and interpret short-term milestones, theoretical continuous time spent inside each clinical compartment, and the final competing risk distributions at the true asymptotic limit.

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1.3. Significance and Benefits

This research provides substantial insights across both theoretical and practical dimensions:

- Theoretical Contributions: This study enriches the domestic statistical literature in Indonesia by providing a rigorous, reproducible documentation of parametric CTMC workflows applied to complex biomedical datasets, effectively bridges pure Markovian transition theory with applied multi-state biostatistics.
- Practical Applications: The calibrated model delivers intuitive, data-driven prognostic matrices and trajectory visualizations capable of supporting clinical epidemiologists in therapeutic decision-making. Furthermore, the established computational pipeline can be readily adapted to evaluate other multi-phase chronic conditions, such as cardiovascular diseases or oncology registries.

1.4. Scope and Limitations

To preserve analytical focus and maintain mathematical consistency, the boundaries of this study are restricted to the following parameters:

- Data Source Constraints: The analysis is strictly secondary, utilizing the historical `ebmt4` bone marrow transplantation registry available within the `mstate` package, which encapsulates the longitudinal records of 2,279 leukemia patients [11]. No primary field data collection was conducted.
- Time-Homogeneity Assumption: The model operates under a strict time-homogeneous constraint where transition intensities are assumed constant over time. While true biological hazards are inherently non-homogeneous, this assumption is enforced as a foundational first-order mathematical approximation.
- Parametric Methodological Focus: Parameter calibration is restricted to parametric Maximum Likelihood Estimation (MLE) for the generator matrix and direct matrix exponential evaluations $\exp(\mathbf{Q}t)$ for transient projections. Non-parametric extensions and semi-parametric regression structures utilizing patient-specific covariates are outside the scope of this study.
- Transition Graph Architecture: The state space is bounded by a 6-state multi-state topology, consisting of four transient phases (*Transplant, Recovery, Adverse Event, and Combined Rec+AE*) and two competing terminal absorbing destinations (*Relapse and Deceased*). Alternative branching paths or re-entry geometries are not explored.

2. METHODOLOGY

This section outlines the mathematical framework of the multi-state model using a continuous-time stochastic approach. The state space configuration, parameter estimation via maximum likelihood, and long-term limiting behaviors are defined to establish the analytical framework.

2.1. Model Description

To model the longitudinal transitions of individuals across distinct clinical phases, a Continuous-Time Markov Chain (CTMC) framework is utilized [5, 14]. Let $\{X(t), t \geq 0\}$ be a continuous-time stochastic process taking values in a finite state space $\mathcal{S} = \{1, 2, \ldots, K\}$. The process satisfies the Markov property for a time-homogeneous process, meaning that for any states $i, j \in \mathcal{S}$ and time points $0 \leq s \leq t$, the conditional probability distribution of any future state depends strictly on the current state:

$$P(X(t) = j \mid X(s) = i, \{X(u) : 0 \leq u < s\}) = P(X(t) = j \mid X(s) = i) \tag{1}$$

The time-homogeneous assumption states that transition intensities depend strictly on the elapsed time between states rather than the absolute historical timeline, serving as an analytical baseline for evaluating transition behavior [1].

2.2. State Space and Transition Architecture

The structural architecture is configured as an interleaved 6-state multi-state system, preserving the natural clinical evolution of individuals post-intervention. The finite state space is defined as $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$, representing:

- State 1 (Tx): Initial baseline state (Initial entry).
- State 2 (Rec): Intermediate transient recovery state.
- State 3 (AE): Intermediate transient adverse event state.
- State 4 (Rec+AE): Combined dual-transient intermediate state.
- State 5 (Rel): Competing absorbing state (Treatment failure via relapse).
- State 6 (Death): Competing absorbing state (Terminal failure).

The permissible instantaneous paths within this system are mathematically restricted. States 5 and 6 are strictly designated as absorbing states, meaning that once the process enters these states, the leaving rate is identically zero ($q_{5j} = q_{6j} = 0, \forall j \in \mathcal{S}$) [6].

2.3. Mathematical Formulation and Intensity Estimation

The transition behavior of the CTMC is governed by the infinitesimal generator matrix, denoted as $\mathbf{Q} \in \mathbb{R}^{K \times K}$ [5]. The off-diagonal elements q_{ij} (where $i \neq j$) represent the instantaneous transition intensities from state i to state j, formulated as:

$$q_{ij} = \lim_{\Delta t \to 0} \frac{P(X(t + \Delta t) = j \mid X(t) = i)}{\Delta t} \tag{2}$$

The diagonal elements q_{ii} represent the total rate of leaving state i and are constrained by the conservation law, forcing the rows of the generator matrix to sum to zero [1]:

$$q_{ii} = -v_i = -\sum_{j \neq i} q_{ij} \tag{3}$$

The parameters of the generator matrix are estimated using parametric Maximum Likelihood Estimation (MLE) under continuous observation [3]. Let D_{ij} denote the total number of observed transitions from state i to state j across the sample. Let E_i represent the cumulative total time-at-risk (total exposure time) that individuals spent inside state i prior to transitioning or censoring. The constant intensity estimator is given by:

$$\hat{q}_{ij} = \frac{D_{ij}}{E_i} = \frac{D_{ij}}{\sum_m (T^{(i)}_{\text{stop},m} - T^{(i)}_{\text{start},m})} \tag{4}$$

2.4. Transient Probabilities and Limiting Behavior

The state probabilities of the system are governed by the transition probability function matrix, $\mathbf{P}(t) \in \mathbb{R}^{K \times K}$, where each entry $P_{ij}(t) = P(X(t) = j \mid X(0) = i)$. The evolution of $\mathbf{P}(t)$ satisfies the system of differential equations described by the Kolmogorov Forward Equations [1]:

$$\frac{d}{dt}\mathbf{P}(t) = \mathbf{P}(t)\mathbf{Q} \tag{5}$$

Under the time-homogeneous assumption, the explicit solution to this system for any duration t is computed via the matrix exponential:

$$\mathbf{P}(t) = \exp(\mathbf{Q}t) = \sum_{k=0}^{\infty} \frac{(\mathbf{Q}t)^k}{k!} \tag{6}$$

The asymptotic characteristics of the system are evaluated by computing the limiting distribution as t approaches infinity. For systems containing absorbing states, the limiting probability matrix captures the ultimate absorption allocation [5]:

$$\mathbf{P}_\infty = \lim_{t\to\infty} \exp(\mathbf{Q}t) \tag{7}$$

2.5. Sojourn Time Distribution

The continuous duration within each transient state is evaluated through the concept of sojourn times. In a time-homogeneous CTMC, the random variable S_i, representing the continuous time an individual resides in a transient state i before transitioning, follows an Exponential distribution parametrized by the leaving rate $v_i = -q_{ii}$ [5]:

$$S_i \sim \text{Exponential}(v_i) \tag{8}$$

Consequently, the expected sojourn time spent inside a specific compartment i is defined as the inverse of its leaving hazard:

$$\mathbb{E}[S_i] = \frac{1}{v_i} = \frac{1}{-q_{ii}} \tag{9}$$

For absorbing states where $q_{ii} = 0$, the expected sojourn time resolves to infinity (∞), representing the mathematical property of an absorbing state.

2.6. Computational Implementation and Software

Data processing and numerical computations are executed within the `R` statistical programming environment. Longitudinal data preparation, specifically the conversion from wide-format records to multi-state long-format exposures, is performed via the `msprep` function from the `mstate` package [11].

The numerical solution for the matrix exponential $\exp(\mathbf{Q}t)$ is evaluated using the `expm` algorithm provided by the `Matrix` package. Longitudinal trajectories and state probability functions are constructed and exported utilizing the `tidyverse` architecture and `ggplot2` visualization libraries. The complete, reproducible computational pipeline and LaTeX source codes are publicly available at https://github.com/zakizulham/Multi-State-Modeling-After-Bone-Marrow-Transplantation.

3. RESULTS AND DISCUSSION

This section presents the empirical outputs, structural estimations, and long-term prognostic trajectories derived from the multi-state calibration of the `ebmt4` cohort data.

3.1. Presentation of Results

The multi-state transformation of the longitudinal registry mapped a total of 12 active clinical transition pathways across 2,279 patients. Table 1 summarizes the empirical event counts and total time-at-risk exposures (expressed in person-days) that serve as the mathematical foundation for the maximum likelihood estimation of the daily transition intensities (q_{ij}).

Table 1. Empirical Transition Events and Exposures for the 6-State Model

From State (i)	To State (j)	Total Events	Total Exposure (Days)	Intensity Rate (q_{ij})
Tx	Rec	785	911,046	0.000862
Tx	AE	907	911,046	0.000996
Tx	Rel	95	911,046	0.000104
Tx	Death	160	911,046	0.000176
Rec	Rec+AE	227	977,570	0.000232
Rec	Rel	112	977,570	0.000115
Rec	Death	39	977,570	0.000040
AE	Rec+AE	433	765,784	0.000565
AE	Rel	56	765,784	0.000073
AE	Death	197	765,784	0.000257
Rec+AE	Rel	107	1,077,153	0.000099
Rec+AE	Death	137	1,077,153	0.000127

These estimated hazards establish the off-diagonal parameters of the infinitesimal generator matrix $\mathbf{Q}$. Enforcing the row-stochastic conservation law yields the complete 6×6 operator structured as follows:

$$\mathbf{Q} = \begin{pmatrix} -0.002137 & 0.000862 & 0.000996 & 0.000000 & 0.000104 & 0.000176 \\ 0.000000 & -0.000387 & 0.000000 & 0.000232 & 0.000115 & 0.000040 \\ 0.000000 & 0.000000 & -0.000896 & 0.000565 & 0.000073 & 0.000257 \\ 0.000000 & 0.000000 & 0.000000 & -0.000227 & 0.000099 & 0.000127 \\ 0.000000 & 0.000000 & 0.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 0.000000 & 0.000000 & 0.000000 \end{pmatrix} \tag{10}$$

Evaluating the negative inverse of the main diagonal elements yields the expected clinical sojourn times spent inside each compartment before a patient transfers out, as tabulated in Table 2.

Table 2. Expected Sojourn Times Across Transient Clinical States

Clinical State	Leaving Rate (v_i)	Expected Sojourn (Days)
Tx	0.002137	467.92
Rec	0.000387	2586.16
AE	0.000896	1116.30
Rec+AE	0.000227	4414.56
Rel	0.000000	∞ (Absorbing State)
Death	0.000000	∞ (Absorbing State)

To map macro-level clinical progressions, the transient transition probability matrices $\mathbf{P}(t)$ were resolved via matrix exponential operations. At the short-term clinical landmark ($t = 100$ days), the matrix $\mathbf{P}(100)$ is given by:

$$\mathbf{P}(100) = \begin{pmatrix} 0.8076 & 0.0760 & 0.0856 & 0.0034 & 0.0102 & 0.0171 \\ 0.0000 & 0.9621 & 0.0000 & 0.0225 & 0.0114 & 0.0041 \\ 0.0000 & 0.0000 & 0.9143 & 0.0535 & 0.0073 & 0.0250 \\ 0.0000 & 0.0000 & 0.0000 & 0.9776 & 0.0098 & 0.0126 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 \end{pmatrix} \tag{11}$$

At the one-year clinical landmark ($t = 365$ days), capturing long-term patient dynamics, the operational matrix $\mathbf{P}(365)$ resolves to:

$$\mathbf{P}(365) = \begin{pmatrix} 0.4584 & 0.2018 & 0.2107 & 0.0351 & 0.0352 & 0.0588 \\ 0.0000 & 0.8684 & 0.0000 & 0.0758 & 0.0404 & 0.0154 \\ 0.0000 & 0.0000 & 0.7211 & 0.1686 & 0.0260 & 0.0843 \\ 0.0000 & 0.0000 & 0.0000 & 0.9206 & 0.0348 & 0.0446 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 \end{pmatrix} \tag{12}$$

Finally, the asymptotic limiting behavior of the multi-state system as $t \to \infty$ evaluates the true terminal absorption matrix, settling the ultimate competing risk distribution:

$$\mathbf{P}(\infty) = \begin{pmatrix} 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.44140 & 0.55860 \\ 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.55964 & 0.44036 \\ 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.35843 & 0.64157 \\ 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.43852 & 0.56148 \\ 0.00000 & 0.00000 & 0.00000 & 0.00000 & 1.00000 & 0.00000 \\ 0.00000 & 0.00000 & 0.00000 & 0.00000 & 0.00000 & 1.00000 \end{pmatrix} \tag{13}$$

3.2. Interpretation

The empirical parameters computed from the 6-state CTMC framework offer granular insights into post-transplantation kinetics, translating abstract transition intensities into concrete clinical prognosis.

3.2.1. *Analysis of Intensities and Sojourn Times*

The estimated infinitesimal generator matrix indicates that patients in the initial transplantation state (`Tx`) exhibit a higher instantaneous tendency to transition directly toward the Adverse Event phase (`AE`, $q_{13} = 0.000996$) than toward pure Recovery (`Rec`, $q_{12} = 0.000862$). Direct terminal transitions from baseline to Relapse ($q_{15} = 0.000104$) or Death ($q_{16} = 0.000176$) remain minimal, confirming that post-BMT failure routes operate predominantly through intermediate transient pathways.

The baseline leaving rate ($v_1 = 0.002137$) translates into an expected sojourn time of 467.92 days, indicating that patients spend a considerable duration in the primary post-graft phase before developing distinct physiological paths. Within the intermediate transient compartments, the pure Recovery phase (`Rec`) manifests a substantial expected sojourn of 2,586.16 days, with its dominant exit path targeting the compound `Rec+AE` state ($q_{24} = 0.000232$). This highlights that even after establishing hematological recovery, patients remain highly exposed to delayed post-transplant complications.

For the acute Adverse Event phase (`AE`), the propensity to progress into the compound history state `Rec+AE` ($q_{34} = 0.000565$) is significantly higher than immediate failure rates ($q_{35} = 0.000073$, $q_{36} = 0.000257$), outlining a clinical buffer zone where acute toxicities frequently overlap with late-stage recoveries. Notably, the compound `Rec+AE` state exhibits an elevated theoretical sojourn time of 4,414.56 days. Once transitions eventually activate from this dual-history state, the daily hazard of mortality ($q_{46} = 0.000127$) systematically dominates over disease relapse ($q_{45} = 0.000099$), pointing to non-relapse mortality driven by cumulative physiological stress as the primary risk.

3.2.2. *Evolution of Transition Probabilities*

Evaluating the matrix exponential at specific time landmarks isolates the macro-level distribution of the patient cohort across various longitudinal stages.

- Short-Term Milestone ($t = 100$ days): At 100 days post-transplantation, a patient entering at `Tx` retains an 80.76% probability of occupying the initial baseline condition. The immediate short-term occupancy for pure Recovery and Adverse Events stands at 7.60% and 8.56% respectively, while early terminal failure remains suppressed (Rel = 1.02%, Death = 1.71%). This mathematically proves that major multi-state differentiation occurs post-landmark.
- One-Year Milestone ($t = 365$ days): After one year, the baseline probability drops substantially to 45.84%. Simultaneously, the probability mass shifts toward active intermediate transient states, with `Rec` and `AE` climbing to 20.18% and 21.07% respectively. Cumulative mortality reaches 5.88%, surpassing relapse occupancy (3.52%), indicating a steady, gradual structural progression.
- Asymptotic Limiting Behavior ($t \to \infty$): At the true asymptotic limit, all transient occupancy probabilities converge to zero, and the cohort is fully absorbed by the two terminal sinks. For a baseline patient (`Tx`), the final distribution converges to a 44.14% probability of ultimate absorption into `Relapse` and a dominant 55.86% probability of absorption into `Death`, resolving the definitive long-term competing risk layout.

The continuous trajectory of these evolving state occupancy functions is visually mapped across a 2,000-day tracking horizon in Figure 1, demonstrating the rapid exhaustion of the baseline phase, the transient peaking of intermediate stages, and the monotonic growth of competing absorbing endpoints.

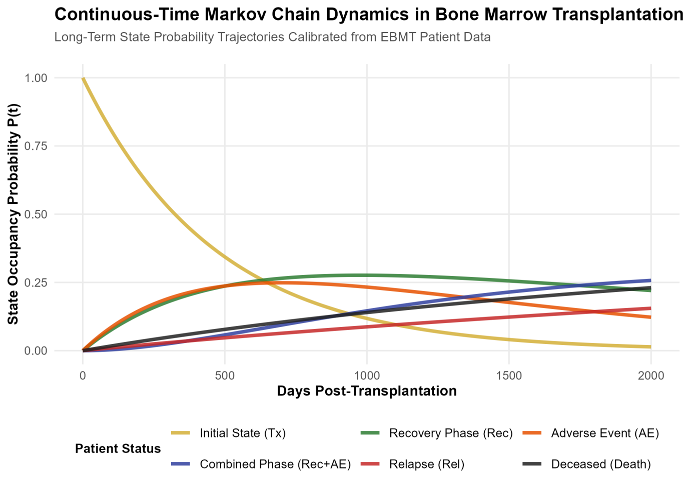

Figure 1. Continuous-Time Markov Chain State Probability Trajectories over 2,000 Days.

3.3. Comparison and Validation

The 6-state CTMC framework provides higher analytical granularity compared to standard non-parametric survival tools like the Kaplan-Meier product-limit estimator. Traditional survival curves are mathematically restricted to single-event layouts, tracking cohorts from baseline exclusively to a binary endpoint (overall survival). In such rigid architectures, critical intermediate multi-state trajectories—such as recovering blood counts or suffering acute toxicities—are either completely missed or treated as non-informative right-censoring, which simplifies the longitudinal clinical timeline.

In contrast, the implemented CTMC model validates its prognostic power by simultaneously tracking interlocking intermediate paths and terminal endpoints via $\mathbf{P}(t) = \exp(\mathbf{Q}t)$. For instance, at the one-year mark, a standard Kaplan-Meier curve would only report a blanket survival rate. The multi-state model, however, dissects this survival mass into precise components: proving that 45.84% remain stable at baseline, while 20.18% and 21.07% are currently battling localized recovery and complication states respectively.

Furthermore, the limiting distribution $\mathbf{P}(\infty)$ uncovers an essential clinical phenomenon. Patients who successfully transition into the pure Recovery phase (`Rec`) display a higher final probability of absorbing into Relapse (55.96%) because they survived early procedural mortality risks. Conversely, falling into an Adverse Event (`AE`) increases the final probability of absorption into Deceased (64.16%), providing a mathematical validation of transplant-related toxicities driving non-relapse mortality. This deep prognostic layer, illustrating how past transient status influences final terminal risks, cannot be directly extracted via conventional bivariate survival curves.

3.4. Strengths and Limitations

The primary strength of this 6-state configuration is its capacity to map multidirectional post-BMT trajectories through 12 distinct transition pathways, including a compound history state (`Rec+AE`). This framework provides explicit continuous-time occupancy probabilities and expected sojourn durations, offering higher prognostic granularity than standard bivariate survival models.

However, the model operates under three constraints. First, the time-homogeneous assumption treats transition intensities as constant, whereas biological hazards typically fluctuate over time. Second, the first-order Markov property enforces memorylessness, ignoring the actual duration a patient previously spent in prior states. Finally, the boundaries are data-restricted by the `ebmt4` registry, which treats Relapse as a terminal absorbing state and precludes the evaluation of post-relapse mortality.

4. CONCLUSION

4.1. Summary of Findings

This study demonstrated the implementation of a 6-state time-homogeneous Continuous-Time Markov Chain (CTMC) model on the `ebmt4` dataset (2,279 patients) to evaluate longitudinal transitions across transient (`Tx`, `Rec`, `AE`, `Rec+AE`) and competing absorbing states (`Rel`, `Death`). Based on the calibrated generator matrix $\mathbf{Q}$ and matrix exponentials $\exp(\mathbf{Q}t)$, four core empirical insights are established:
(1) Initial-phase hazard profiles: In the `Tx` state, the transition intensity toward Adverse Events ($q_{13} = 0.000996$/day) exceeded that toward

Platelet Recovery ($q_{12} = 0.000862$/day), indicating a higher early susceptibility to complications than to recovery, with an expected baseline sojourn time of 467.92 days.
(2) Sojourn time distribution parameters: Among the transient phases, the combined `Rec+AE` state exhibited the longest expected sojourn (4, 414.56 days), followed by `Rec` (2, 586.16 days) and `AE` (1, 116.30 days). In the `Rec+AE` compartment, mortality constituted the dominant risk as the daily hazard of Death ($q_{46} = 0.000127$) systematically exceeded Relapse ($q_{45} = 0.000099$).
(3) Time-specific probability landmarks: At 100 days post-BMT, 80.76% of patients starting in `Tx` remained in the initial baseline condition. By 365 days, this occupancy dropped to 45.84%, while `Rec` and `AE` states rose to 20.18% and 21.07% respectively, confirming a gradual structural transition layout.
(4) Asymptotic long-run absorption: As $t \to \infty$, the state dynamics for a baseline patient (`Tx`) converge to an ultimate terminal absorption allocation of 44.14% for `Relapse` and 55.86% for `Death`.

Collectively, these multi-state parameters provide a statistically consistent and clinically relevant representation of post-transplantation dynamics that cannot be captured by a standard Kaplan-Meier curve.

4.2. Implications

Methodologically, this work confirms the CTMC multi-state framework as a principled alternative to bivariate survival models. Calibrating 12 transition intensities simultaneously through the MLE estimator D_{ij}/E_i, and resolving the full probability trajectory through the matrix exponential $\exp(\mathbf{Q}t)$, avoids the information loss caused by treating intermediate clinical events as censored data points. The complete analytical workflow, from `msprep()` long-format conversion through matrix exponential computation via `expm` to `ggplot2` visualization, is structured to be reproducible and adaptable to similar longitudinal clinical registries.

On the applied side, the baseline `Tx`-to-`AE` intensity exceeding the `Tx`-to-`Rec` intensity serves as a statistical indicator that clinical complications frequently precede stable recovery for a large portion of the cohort, warranting close monitoring in the early post-transplant window. The dominance of Death over Relapse as the final absorbing state, sitting at 55.86% versus 44.14% for a `Tx`-origin patient, further reinforces that clinical protocols focused on mortality prevention across all transient phases deserve equal or greater attention than relapse prevention alone. These are granular risk-differentiations that standard univariate survival curves cannot extract.

4.3. Recommendations

Several methodological directions are proposed to extend the analytical utility of this framework. *First,* the time-homogeneous assumption constrains the transition intensities q_{ij} to remain constant. In clinical applications, these hazards typically exhibit time-dependent fluctuations, often peaking in the acute post-transplantation phase. Future research should examine time-inhomogeneous CTMC specifications or semi-Markov models to permit state-specific empirical holding time distributions outside the Exponential family.

Second, the first-order Markov property assumes future transition probabilities are independent of the historical duration spent in prior states. Incorporating covariate-stratified Cox proportional hazards extensions within the `mstate` framework—linking patient-level variables such as age, disease subtype, and conditioning regimen directly to the intensity estimates—would address this memoryless constraint and refine the estimation of the generator matrix $\mathbf{Q}$.

Third, the prognostic scope is bounded by the tracking limitations of the `ebmt4` dataset, which classifies Relapse as a terminal absorbing state alongside Death. This structural constraint precludes the evaluation of post-relapse mortality dynamics. Utilizing extended longitudinal registries with systematic post-recurrence tracking would enable the mathematical modeling of these subsequent transition pathways.

Fourth, future work should incorporate the non-parametric Aalen-Johansen estimator as an analytical complement to the maximum likelihood approach, resolving transition probabilities without enforcing time-homogeneity constraints. *Finally,* executing formal Markov property verifications and calibrating this multi-state framework using primary Indonesian clinical data represent essential prerequisites before these stochastic outputs can be applied to inform localized therapeutic decisions.

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